

Full name _Abubakar Bello Sadiq

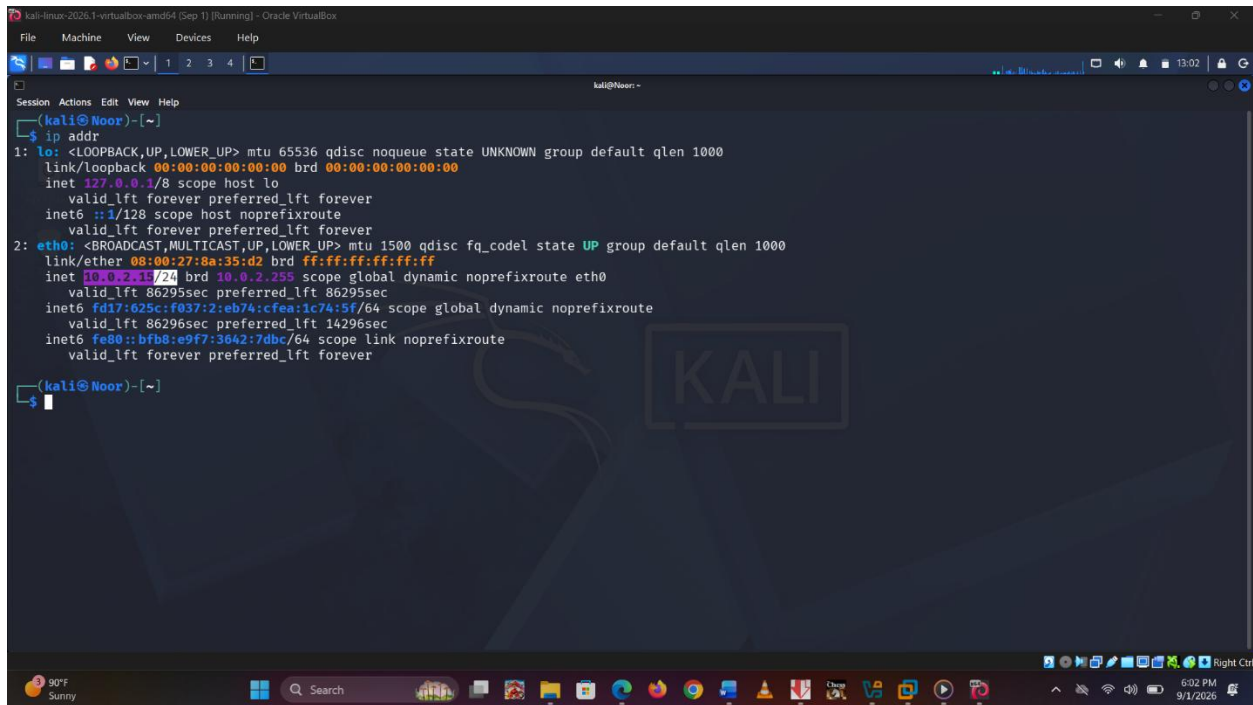
Registration number _C11/26/FCDF/17161

Linux distribution / version_ Linux Kali 6.18.12 + Kali-amd64

VM hostname _ Noor

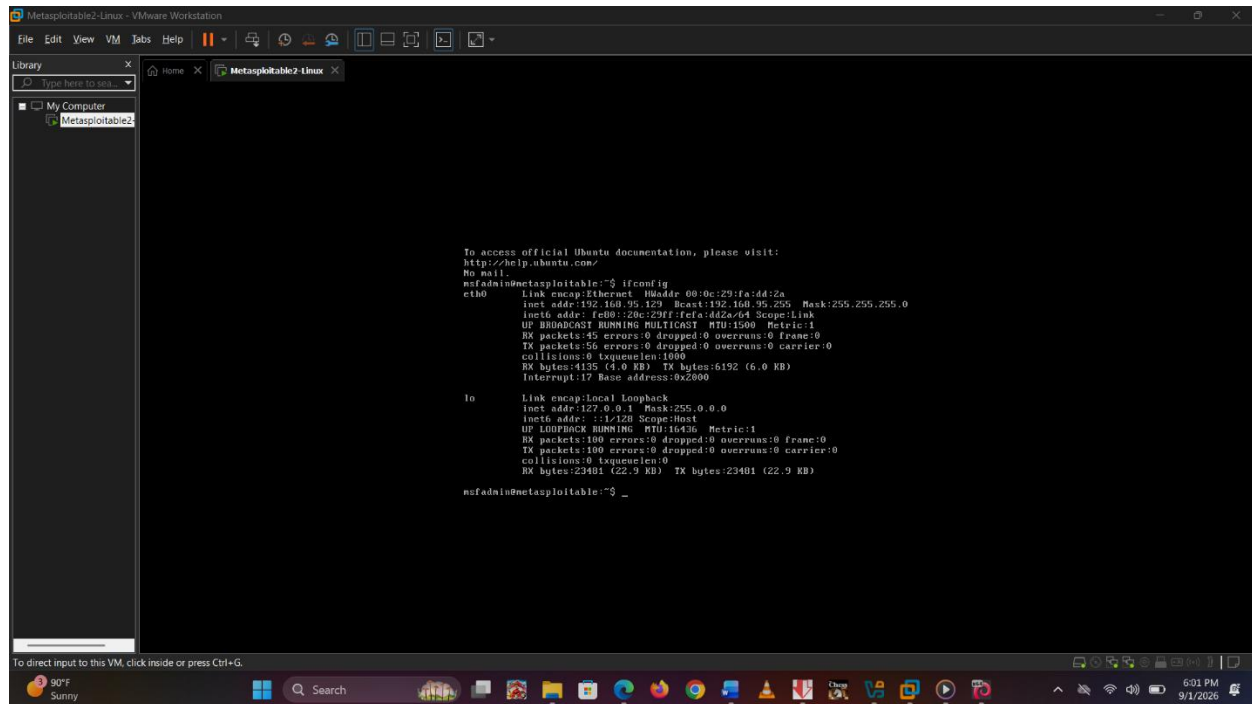
Lab 1: Network & Service Reconnaissance with Metasploitable 2

Step 1: Find Kali IP

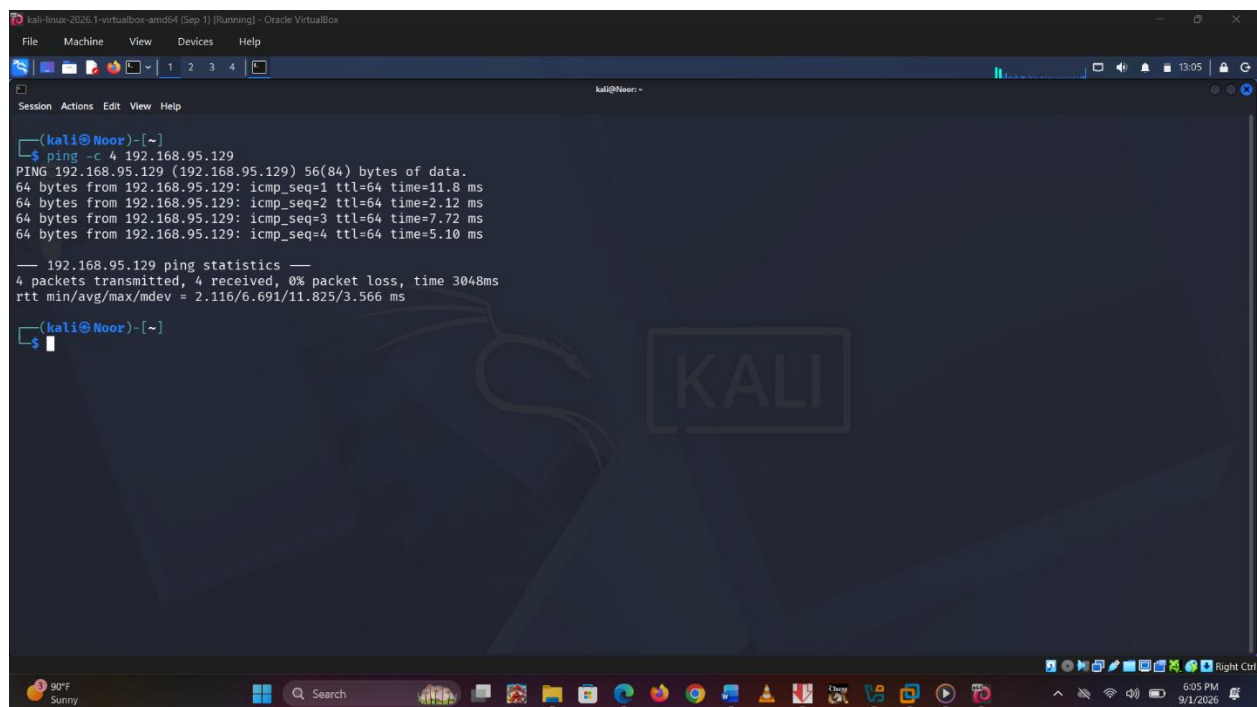


```
kali@Noor: ~  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86295sec preferred_lft 86295sec  
    inet6 fd17:025c:f037:2:eb74:cfea:1c74:5f/64 scope global dynamic noprefixroute  
        valid_lft 86296sec preferred_lft 14296sec  
    inet6 fe80::bfb8:e9f7:3642:7db< /64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
kali@Noor: ~  
$
```

Step 2: Find Metasploitable 2 IP

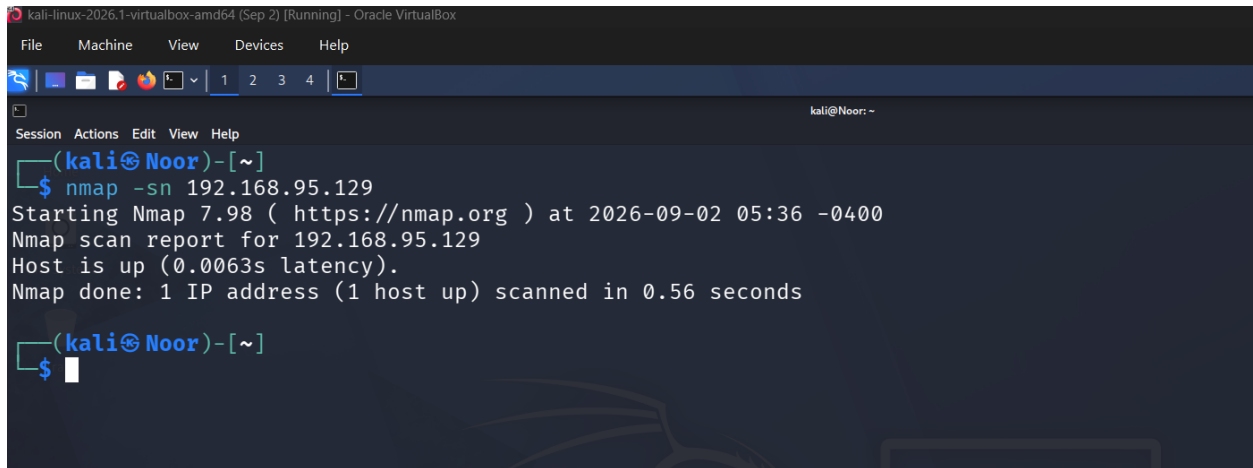


Step 3: Confirm connectivity: This shows that the system is communicating with the other system.



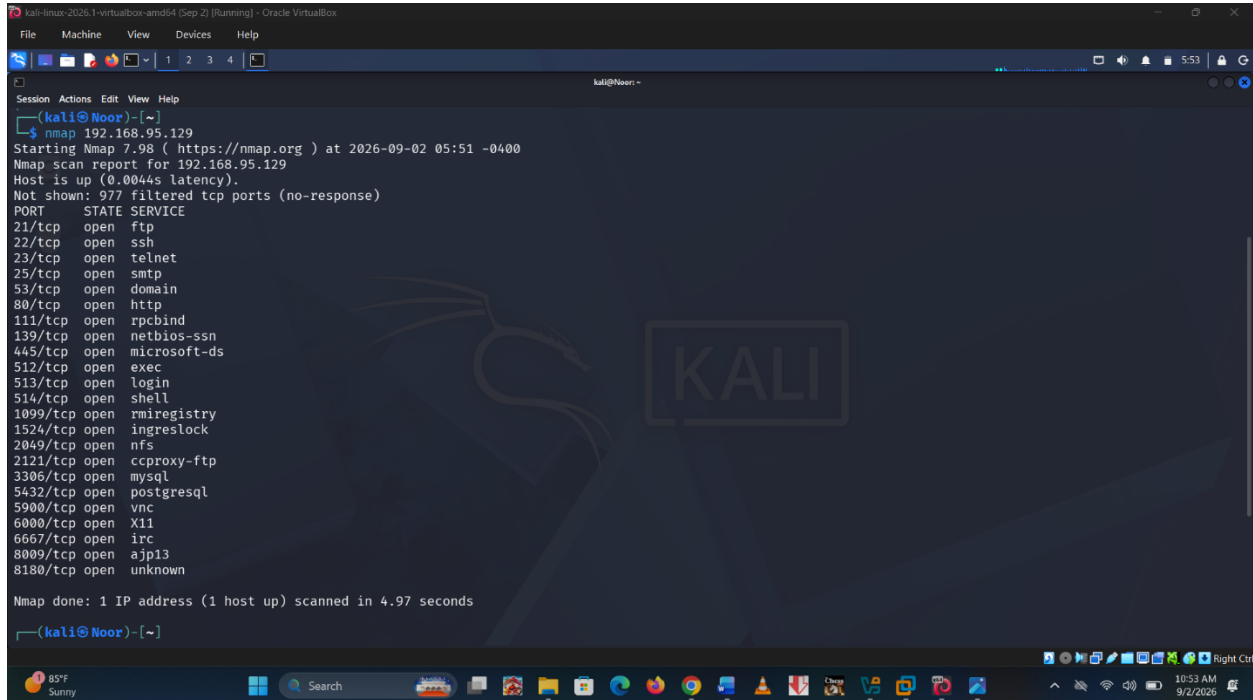
Part 3 - Core Nmap Scans

Step 4: Host discovery: I used (nmap -sn 192.168.95.129) to strip away the port-scanning phase entirely and force it to only use ICMP Echo requests (pings) as seen. And the I used Metasploitable 2 ip addr



```
kali-linux-2026.1-virtualbox-amd64 (Sep 2) [Running] - Oracle VM VirtualBox
File Machine View Devices Help
1 2 3 4
kali@Noor: ~
Session Actions Edit View Help
(kali@Noor)-[~]
$ nmap -sn 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 05:36 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0063s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds
(kali@Noor)-[~]
$
```

Step 5: Default TCP scan: As seen all open TCP port are shown, which is used (nmap 192.168.95.129) which is basic scanning.

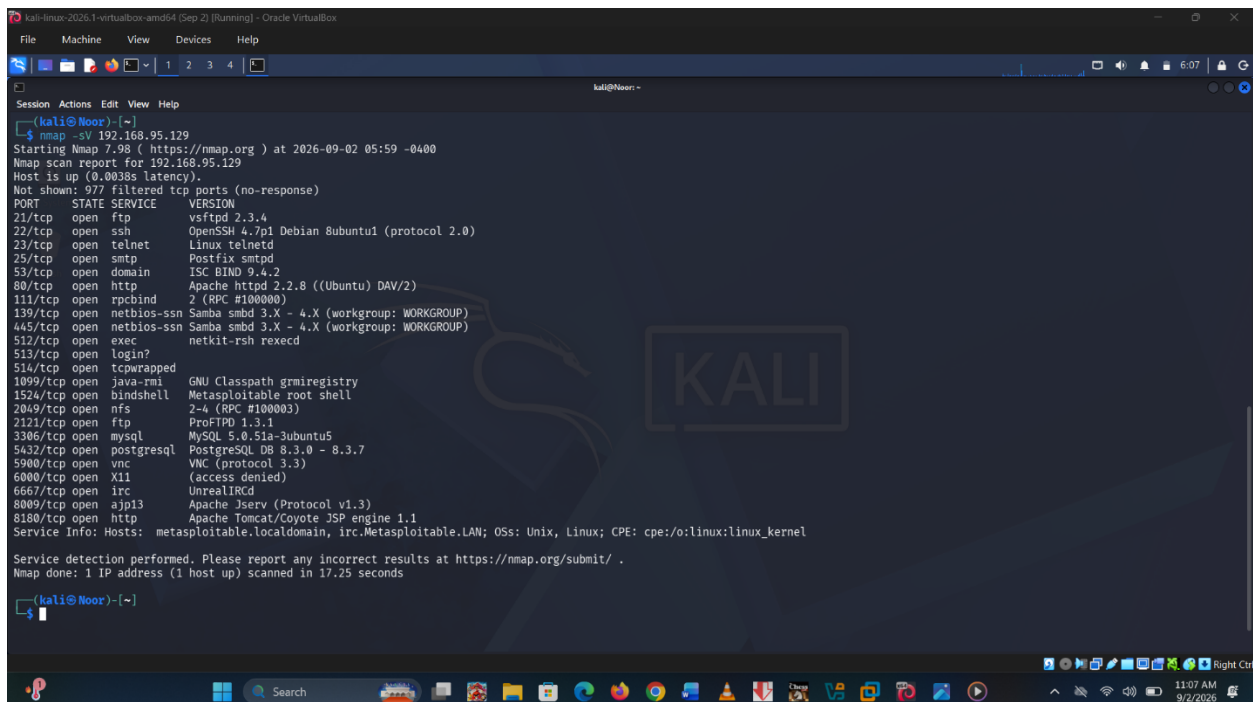


```
kali@kali:~$ nmap 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 05:51 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0044s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 4.97 seconds

kali@kali:~$
```

Step 6: Service/version detection: As I used (nmap -sV 192.168.95.129) to display the product version.



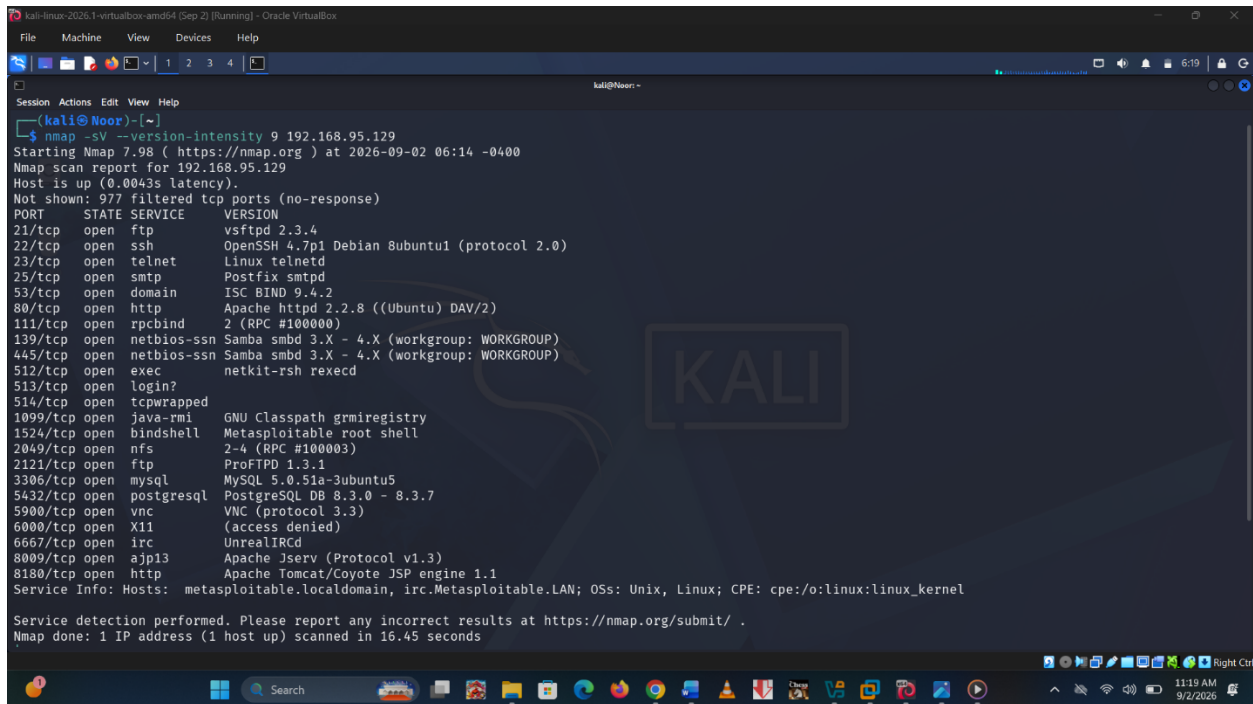
```
kali@kali:~$ nmap -sV 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 05:59 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0038s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1

Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.25 seconds

kali@kali:~$
```

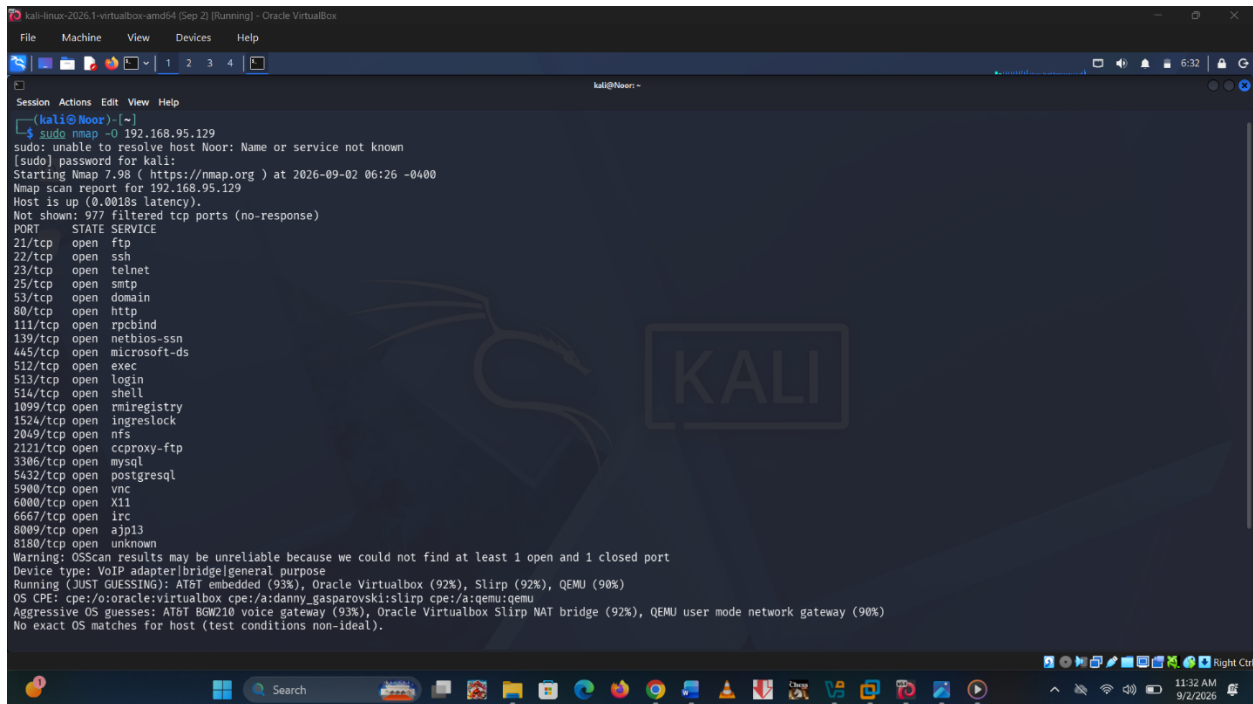
Step 7: Maximum version intensity: Base on the two-result compare (-sV and sV --version-intensity 9) on mine there is no change on the both side of the two result as seen.



```
kali@kali:~$ nmap -sV --version-intensity 9 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 06:14 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0043s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

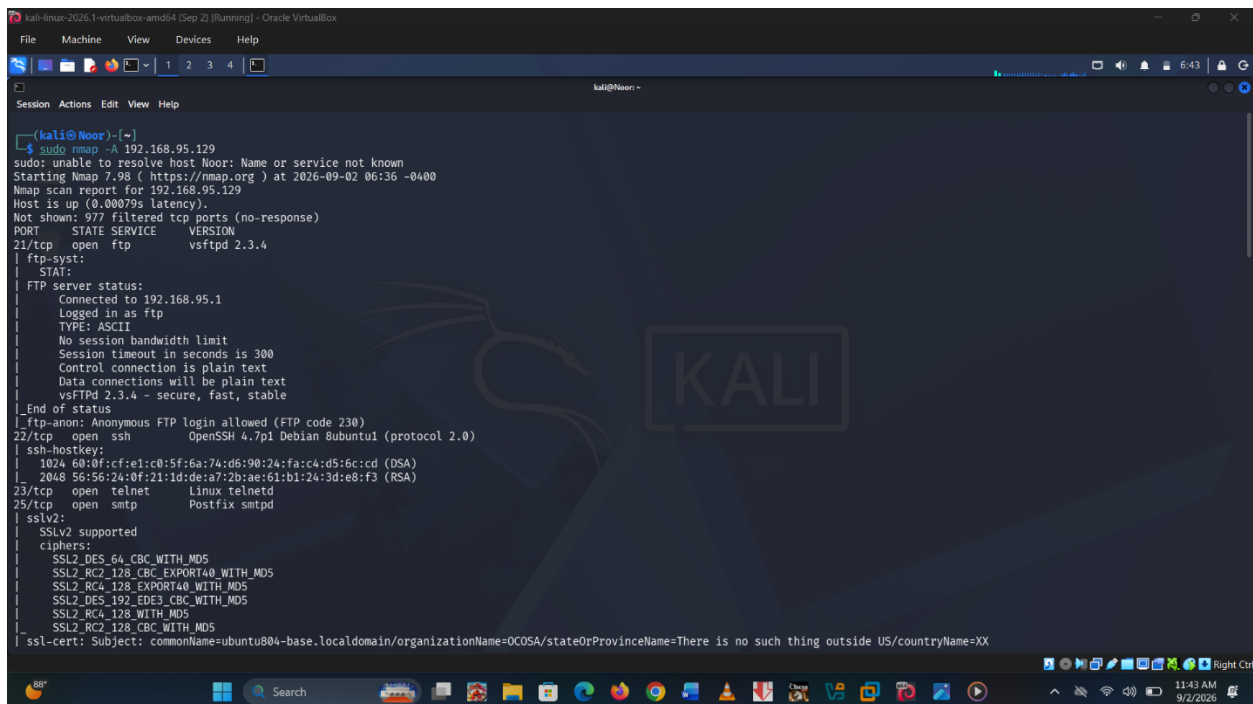
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.45 seconds
```

Step 8: OS detection: It give us the number of ports that open and the details of the information regarding the system by using (sudo nmap -O 192.168.95.129) as seen after using the command.



```
kali@kali:~$ sudo nmap -O 192.168.95.129
sudo: unable to resolve host Noor: Name or service not known
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 06:26 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0018s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmlregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T RCW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
```

Step 9: Aggressive scan: I used (sudo nmap -A 192.168.95.129)



```
kali@kali:~$ sudo nmap -A 192.168.95.129
sudo: unable to resolve host Noor: Name or service not known
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 06:36 -0400
Nmap scan report for 192.168.95.129
Host is up (0.00079s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ ftp-syst:
|_ STAT:
|_ FTP server status:
|_   Connected to 192.168.95.1
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ sslv2:
|_   SSLv2 supported
|_   ciphers:
|_     SSL2_DES_64_CBC_WITH_MD5
|_     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_     SSL2_RC4_128_EXPORT40_WITH_MD5
|_     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_     SSL2_RC4_128_WITH_MD5
|_     SSL2_RC2_128_CBC_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
```

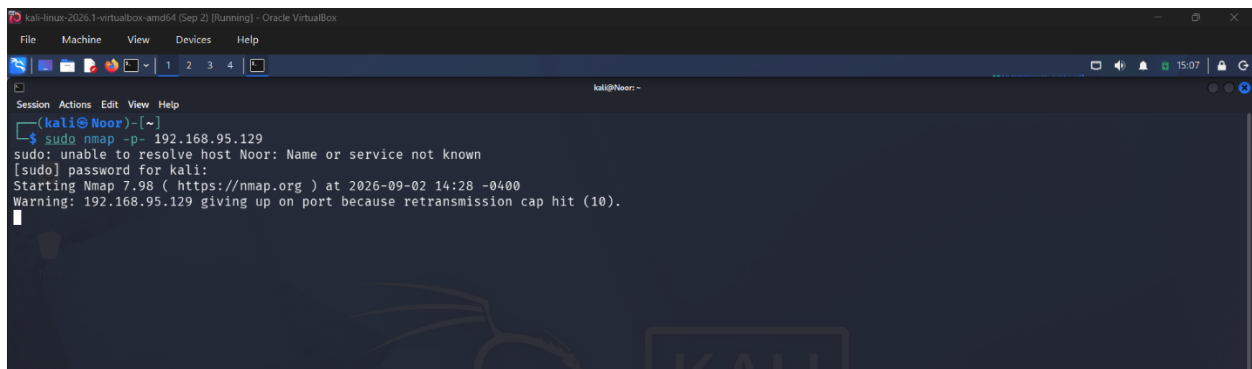
The comparison between the Step 5 default scan and Step 9 aggressive scan

Feature	Default Scan	Aggressive Scan
Speed	Fast (seconds)	Slower (can take more)
OS Fingerprint	No	Yes (Extracts exact banners)
Vulnerability Scripts	No	Yes

Step 10: Full TCP port scan

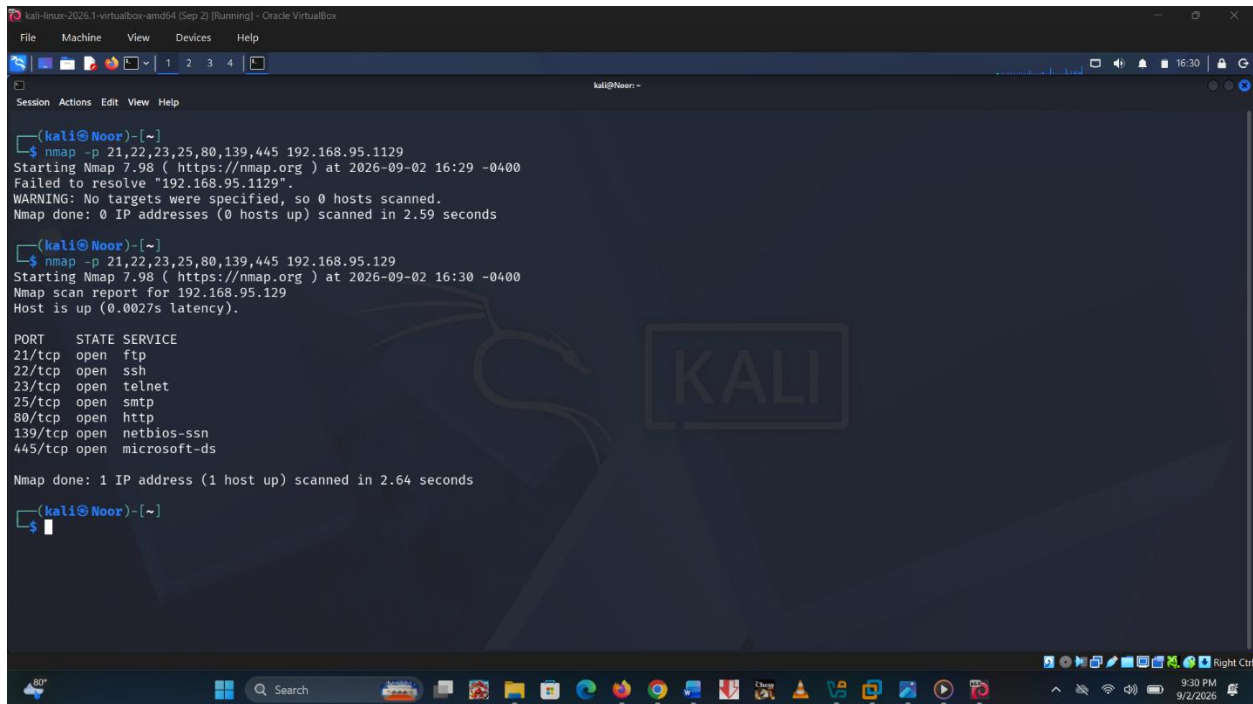
Step 10, 11 and 12 all steps ports are giving up because retransmission cap hit ether 10 or 6.

This is the result I am receiving from the terminal.

A screenshot of a terminal window titled 'kali@Noor: ~'. The terminal shows the command 'sudo nmap -p- 192.168.95.129' being executed. The output indicates that the host 'Noor' cannot be resolved, a password was entered for 'kali', and Nmap 7.98 started at 2026-09-02 14:28 -0400. A warning message states: 'Warning: 192.168.95.129 giving up on port because retransmission cap hit (10)'. The terminal background features a faint Kali Linux logo.

```
kali@Noor: ~  
$ sudo nmap -p- 192.168.95.129  
sudo: unable to resolve host Noor: Name or service not known  
[sudo] password for kali:  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 14:28 -0400  
Warning: 192.168.95.129 giving up on port because retransmission cap hit (10).  
^
```


Step 13: Selected ports: when running (nmap -p) which only selected ports are scanning, I wrongly used wrong ip addrs as seen before change to correct ip addrs. And all the ports are open as seen.



```
(kali@Noor)-[~]
$ nmap -p 21,22,23,25,80,139,445 192.168.95.1129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 16:29 -0400
Failed to resolve "192.168.95.1129".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 2.59 seconds

(kali@Noor)-[~]
$ nmap -p 21,22,23,25,80,139,445 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 16:30 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0027s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds

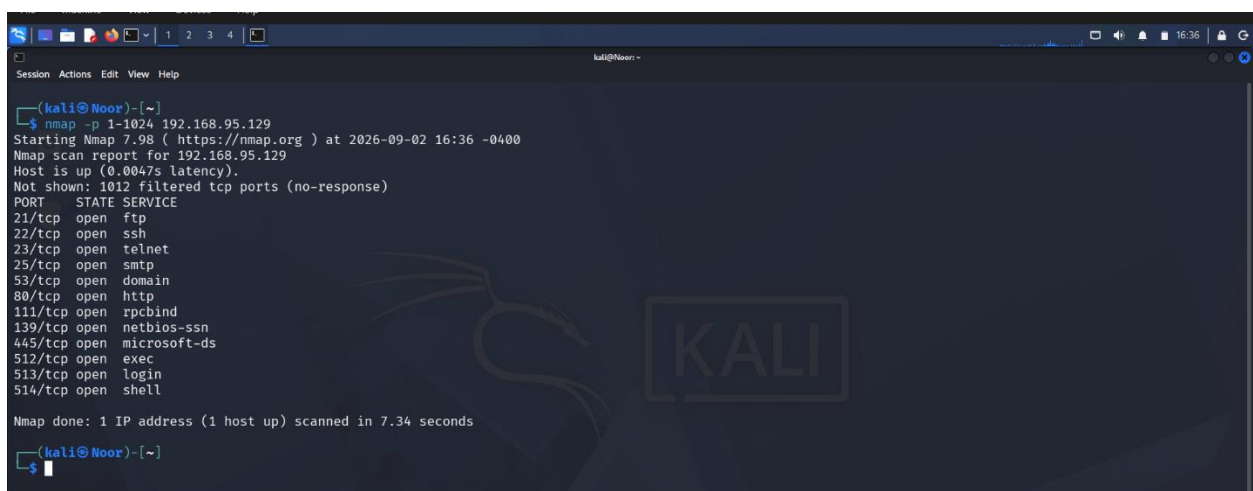
Nmap done: 1 IP address (1 host up) scanned in 2.64 seconds

(kali@Noor)-[~]
$
```

Step 14: Port range: As this shows that scanning from port 1 to port 1024 as seen.

Step 13 and Step 14 have different specific instructions on them.

Step 13 instructs Nmap to scan only seven specific ports on a target, while Step 14 specific on port ranging from port 1 to port 1024 a sign to them as seen.

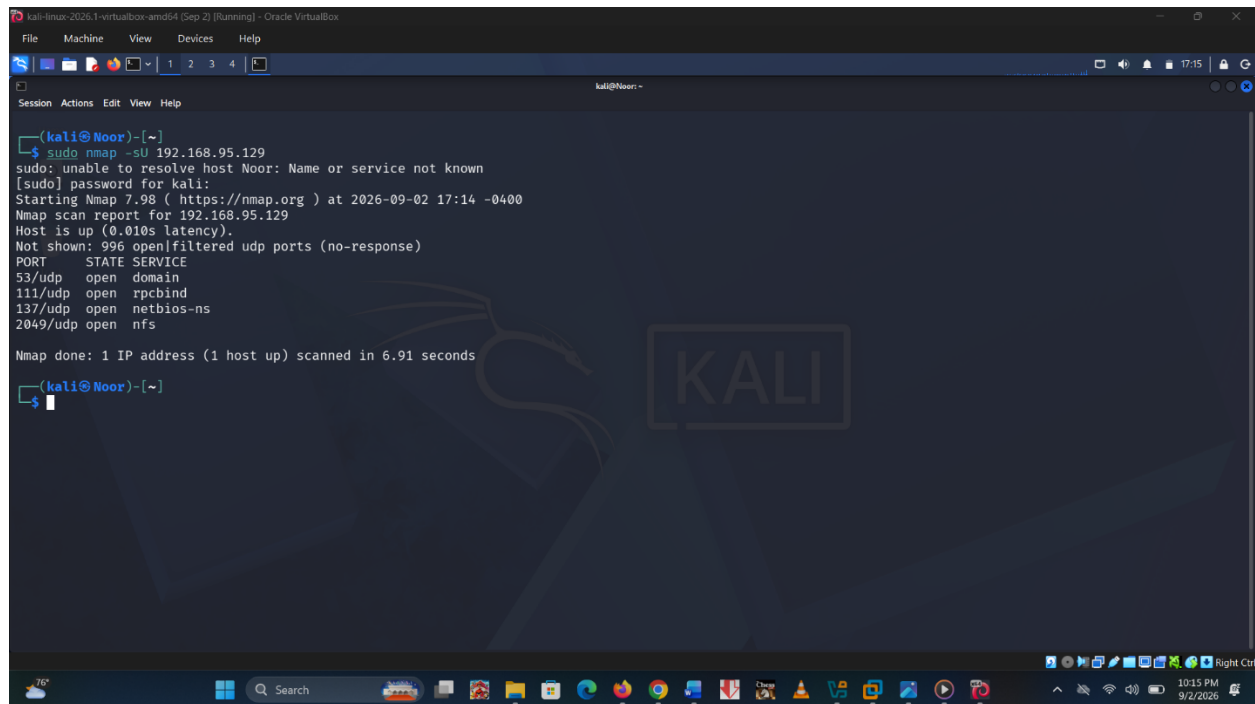


```
(kali@Noor)-[~]
$ nmap -p 1-1024 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 16:36 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0047s latency).
Not shown: 1012 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell

Nmap done: 1 IP address (1 host up) scanned in 7.34 seconds

(kali@Noor)-[~]
$
```

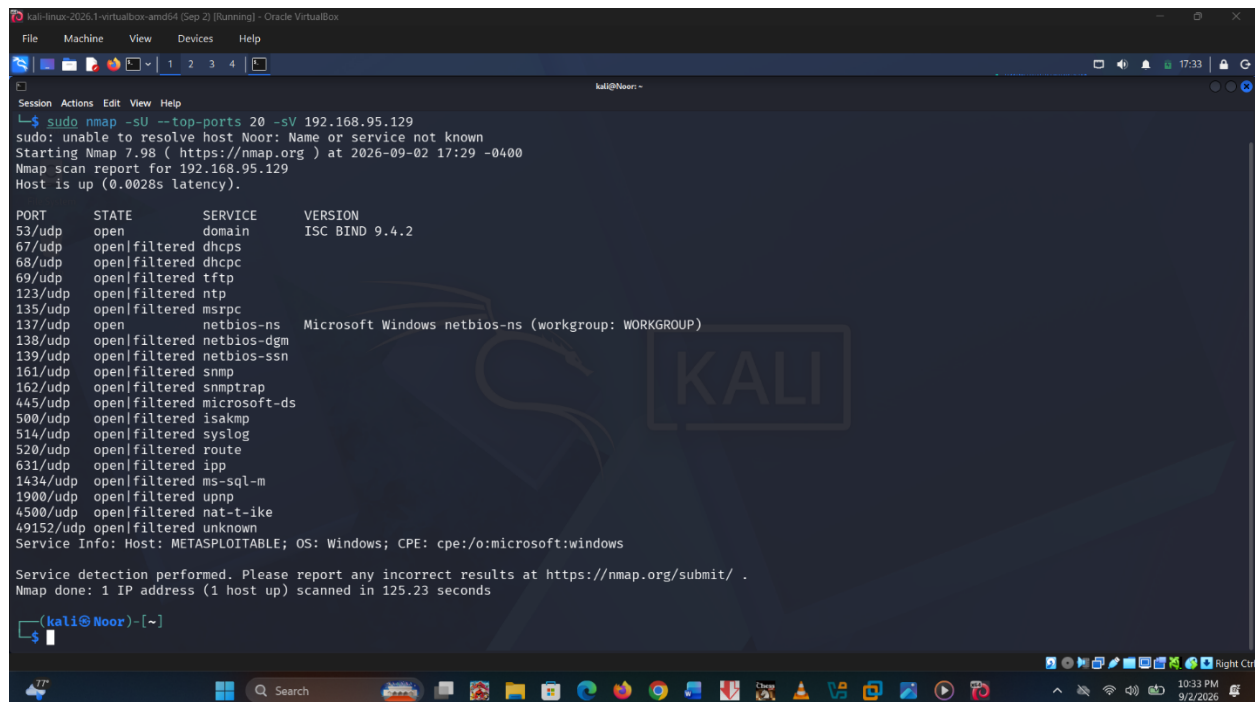

Step 15: UDP scan: Instructs Nmap to check for open User Datagram Protocol (UDP) ports as seen, which I used (sudo nmap -sU)



```
(kali@Noor)~$ sudo nmap -sU 192.168.95.129
sudo: unable to resolve host Noor: Name or service not known
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 17:14 -0400
Nmap scan report for 192.168.95.129
Host is up (0.010s latency).
Not shown: 996 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
137/udp   open  netbios-ns
2049/udp  open  nfs

Nmap done: 1 IP address (1 host up) scanned in 6.91 seconds
(kali@Noor)~$
```

Step 16: Top UDP ports + versions: As seen the Scans with a small common UDP set and identifies services



```
(kali@Noor)~$ sudo nmap -sU --top-ports 20 -sV 192.168.95.129
sudo: unable to resolve host Noor: Name or service not known
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 17:29 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0028s latency).

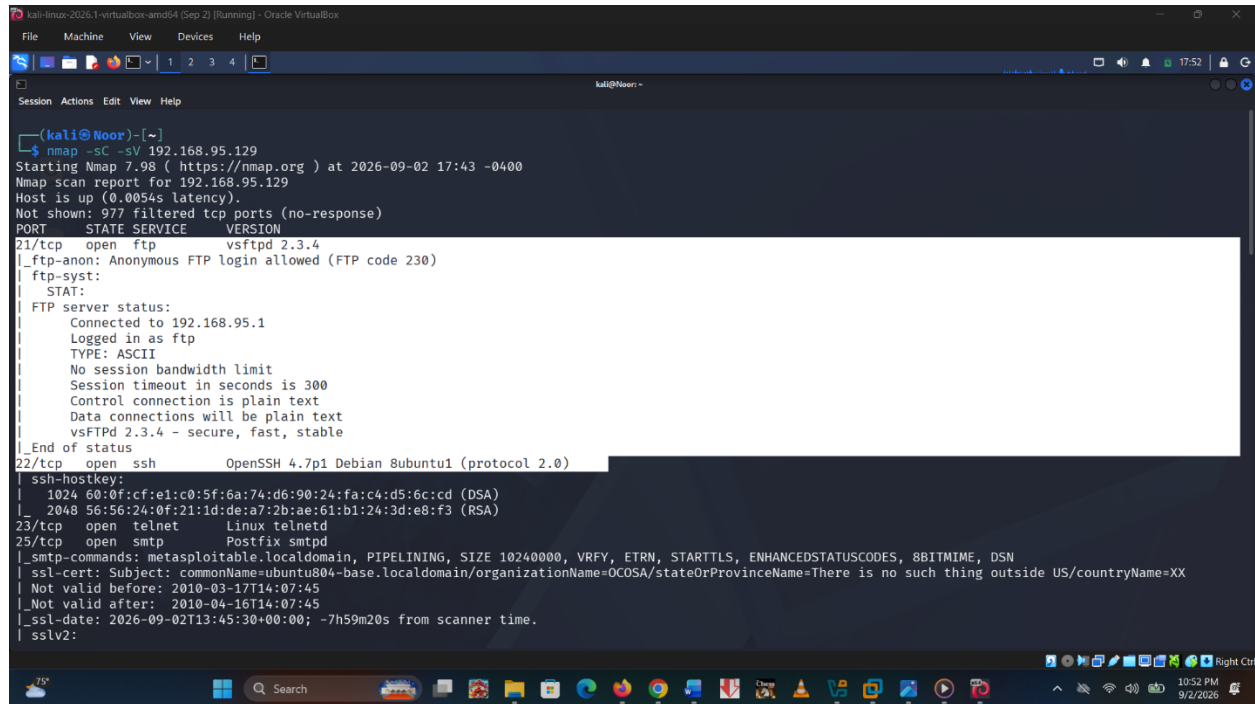
PORT      STATE SERVICE      VERSION
53/udp    open  domain       ISC BIND 9.4.2
67/udp    open|filtered dhcpd        ISC DHCP 3.12.2
68/udp    open|filtered dhcpd        ISC DHCP 3.12.2
69/udp    open|filtered tftp        tftpd 0.25.0
123/udp   open|filtered ntp         ntpd 4.2.6
135/udp   open|filtered msrpc       Microsoft Windows netbios-ns (workgroup: WORKGROUP)
137/udp   open  netbios-ns   Microsoft Windows netbios-ns (workgroup: WORKGROUP)
138/udp   open|filtered netbios-dgm  Microsoft Windows netbios-dgm (workgroup: WORKGROUP)
139/udp   open|filtered netbios-ssn  Microsoft Windows netbios-ssn (workgroup: WORKGROUP)
161/udp   open|filtered snmp        Net-SNMP 5.9.1
162/udp   open|filtered snmptrap    Net-SNMP 5.9.1
445/udp   open|filtered microsoft-ds Microsoft Windows [unidentified] (workgroup: WORKGROUP)
500/udp   open|filtered isakmp      Internet Security Association and Key Management Protocol
514/udp   open|filtered syslog     syslogd 5.5.1
520/udp   open|filtered route      routed 5.1
631/udp   open|filtered ipp         IETF IPP 2.0
1434/udp  open|filtered ms-sql-m    Microsoft SQL Server 2008 R2
1900/udp  open|filtered upnp       Universal Plug and Play
4500/udp  open|filtered nat-t-ike  Microsoft Windows [unidentified] (workgroup: WORKGROUP)
49152/udp open|filtered unknown    unknown

Service Info: Host: METASPLOITABLE; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 125.23 seconds
(kali@Noor)~$
```

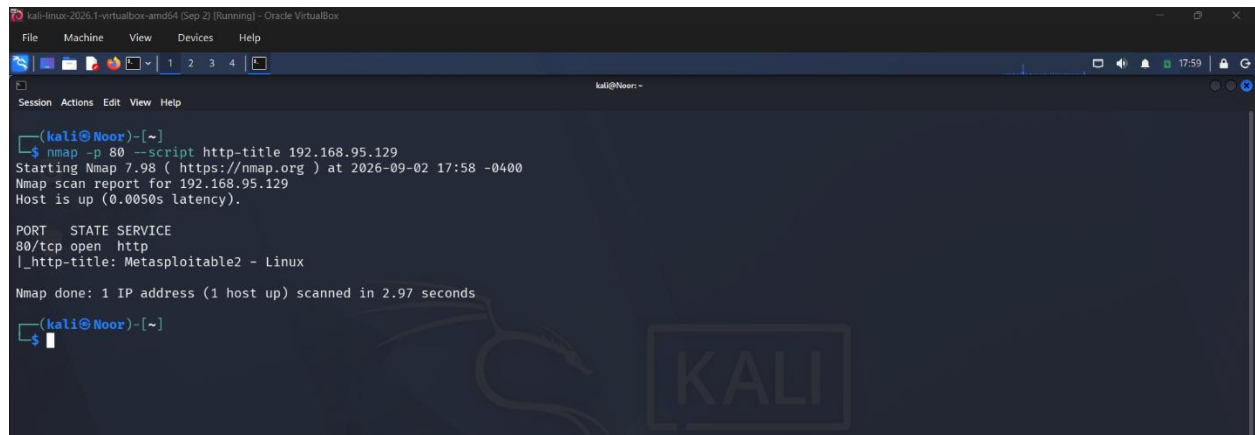
Part 4 - Safe Nmap NSE Enumeration

Step 17: Default scripts: As seen the two useful scripts have



```
(kali@Noor)-[~]
$ nmap -sC -sV 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 17:43 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0054s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
ftp-syst:
STAT:
FTP server status:
  Connected to 192.168.95.1
  Logged in as ftp
  TYPE: ASCII
  No session bandwidth limit
  Session timeout in seconds is 300
  Control connection is plain text
  Data connections will be plain text
  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_ 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-09-02T13:45:30+00:00; -7h59m20s from scanner time.
|_sslv2:
```

Step 18: HTTP title: This is what the result shows

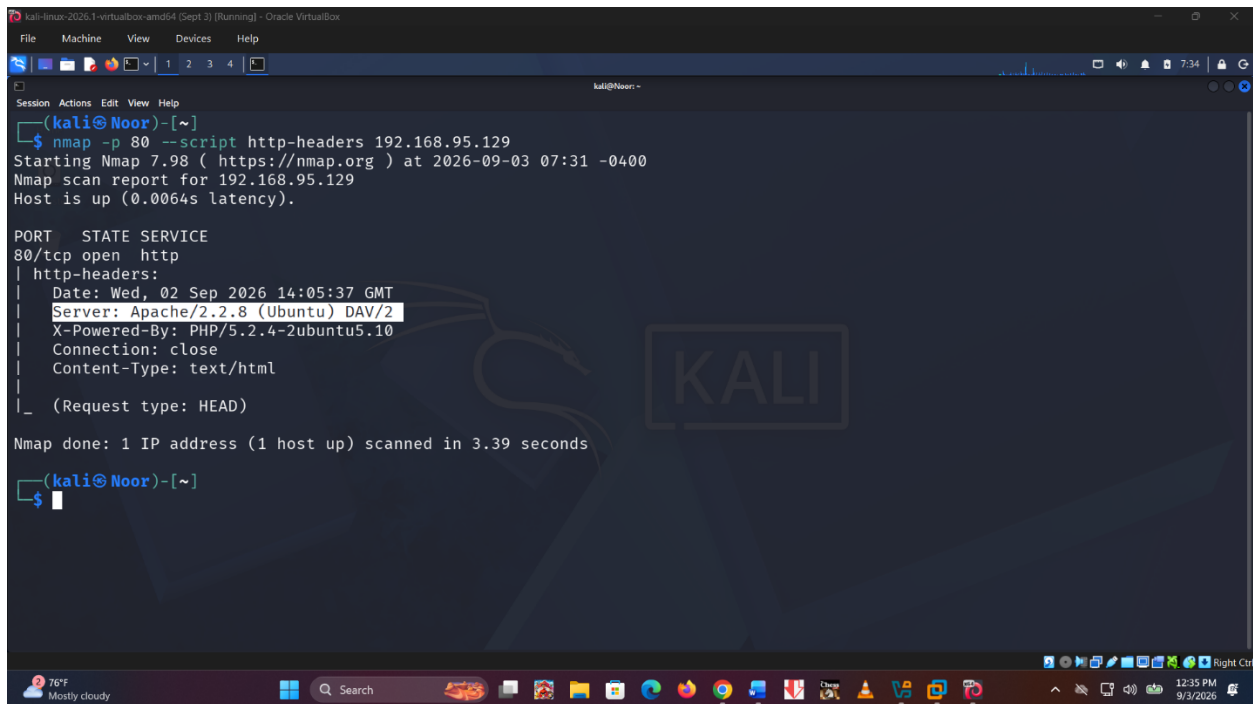


```
(kali@Noor)-[~]
$ nmap -p 80 --script http-title 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 17:58 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0050s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_http-title: Metasploitable2 - Linux

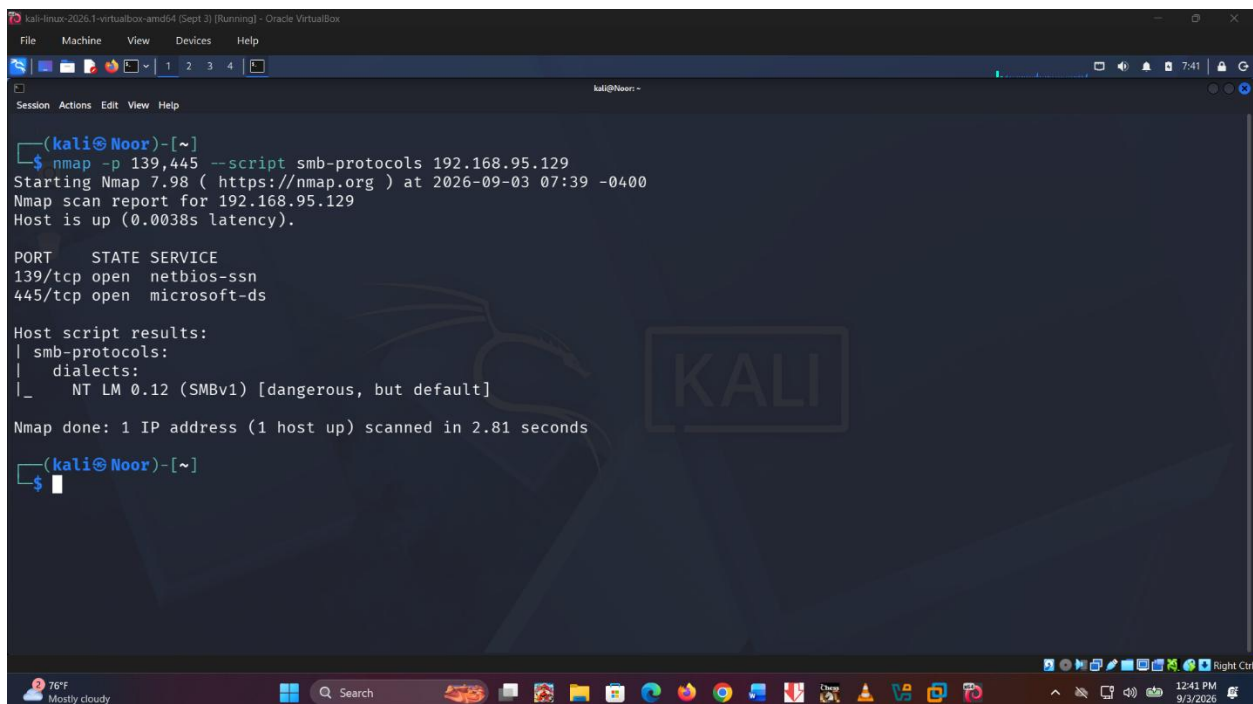
Nmap done: 1 IP address (1 host up) scanned in 2.97 seconds
(kali@Noor)-[~]
$
```

Step 19: HTTP headers: After the command is executed, I highlighted server header



```
kali@Noor: ~  
$ nmap -p 80 --script http-headers 192.168.95.129  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-03 07:31 -0400  
Nmap scan report for 192.168.95.129  
Host is up (0.0064s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
| http-headers:  
|   Date: Wed, 02 Sep 2026 14:05:37 GMT  
|   Server: Apache/2.2.8 (Ubuntu) DAV/2  
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10  
|   Connection: close  
|   Content-Type: text/html  
|  
|_ (Request type: HEAD)  
  
Nmap done: 1 IP address (1 host up) scanned in 3.39 seconds  
  
kali@Noor: ~  
$
```

Step 20: SMB protocols: Server Message Block (SMB) is a client-server communication protocol used for sharing files, printers, and serial ports between devices on a network



```
kali@Noor: ~  
$ nmap -p 139,445 --script smb-protocols 192.168.95.129  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-03 07:39 -0400  
Nmap scan report for 192.168.95.129  
Host is up (0.0038s latency).  
  
PORT      STATE SERVICE  
139/tcp    open  netbios-ssn  
445/tcp    open  microsoft-ds  
  
Host script results:  
| smb-protocols:  
|   dialects:  
|_    NT LM 0.12 (SMBv1) [dangerous, but default]  
  
Nmap done: 1 IP address (1 host up) scanned in 2.81 seconds  
  
kali@Noor: ~  
$
```

Step 21: SSH host keys

```
(kali@Noor)-[~]
$ nmap -p 22 --script ssh-hostkey 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-03 07:56 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0012s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)

Nmap done: 1 IP address (1 host up) scanned in 2.86 seconds

(kali@Noor)-[~]
$
```

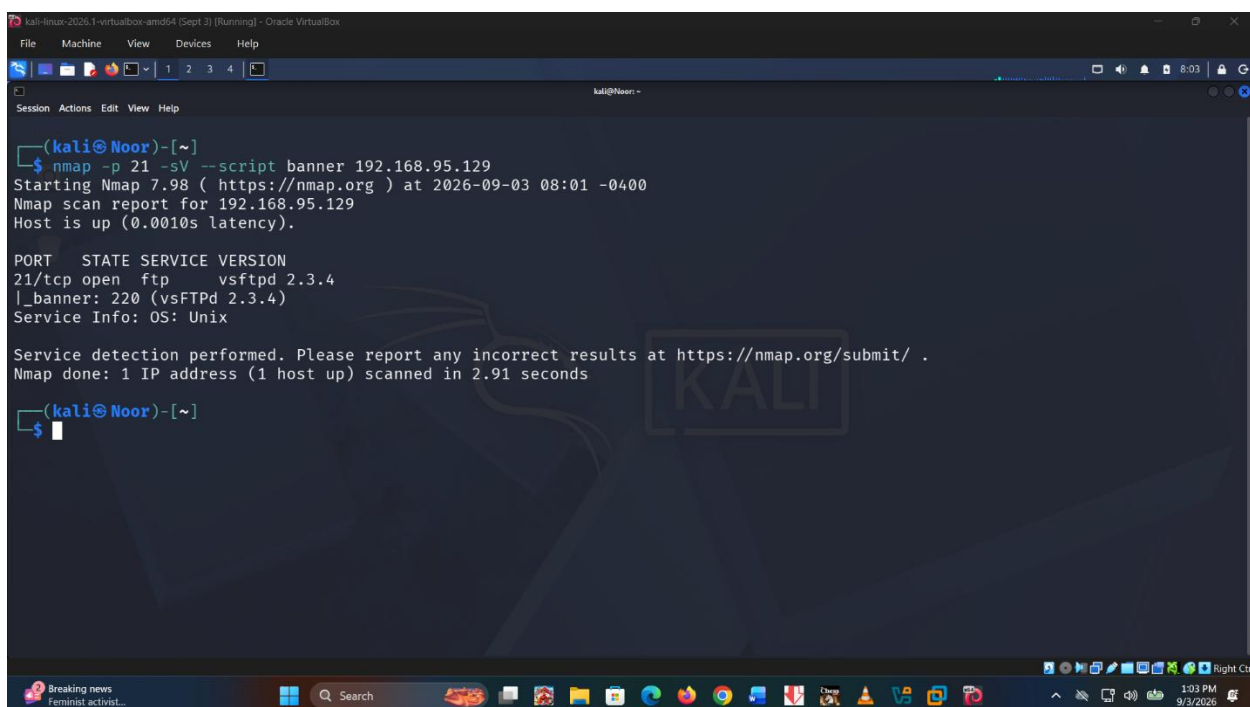
The algorithm names

ssh-hostkey:

1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)

2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)

Step 22: FTP banner



```
kali@Noor: ~
$ nmap -p 21 -sV --script banner 192.168.95.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-03 08:01 -0400
Nmap scan report for 192.168.95.129
Host is up (0.0010s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_ banner: 220 (vsFTPd 2.3.4)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.91 seconds

(kali@Noor)-[~]
$
```

Compare banner and VERSION result.

Banner

220 (vsFTPD 2.3.4)

Low (can easily be faked or spoofed).

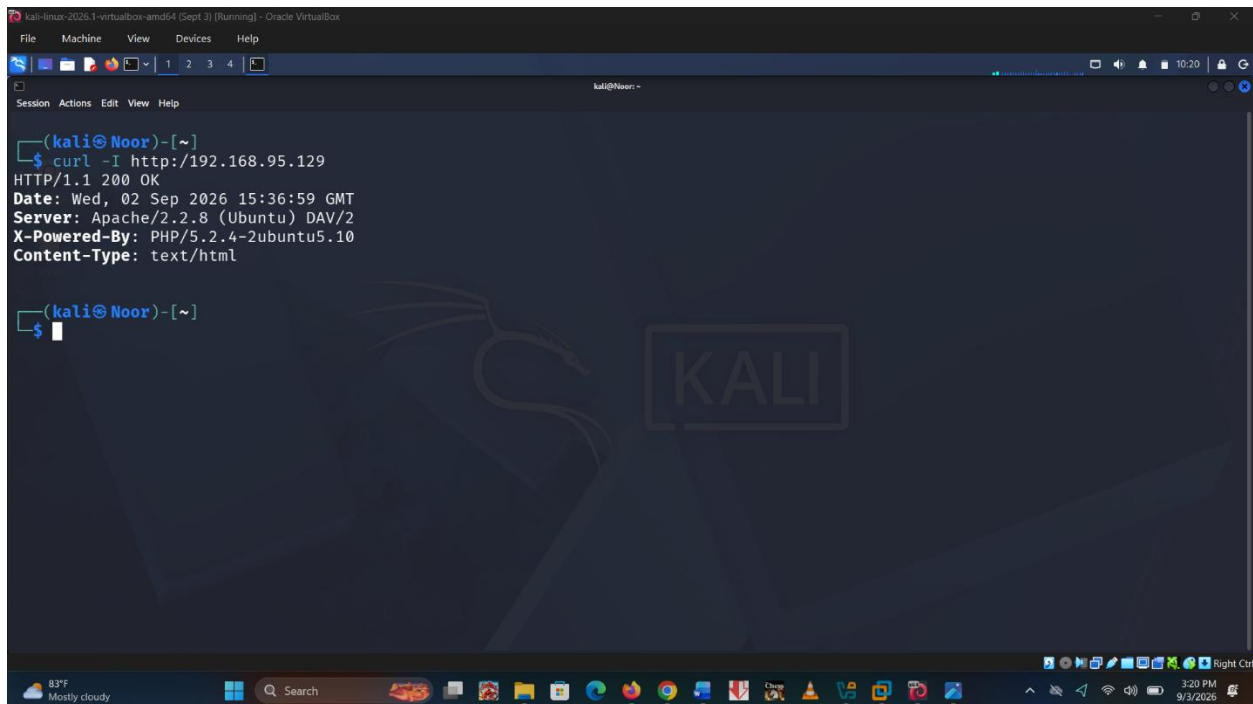
Version

vsftpd 2.3.4.\

High (cross-referenced with signatures).

Part 5 - Manual HTTP Confirmation

Step 23: Request headers with curl

A screenshot of a Kali Linux terminal window. The terminal shows a command prompt where the user has entered 'curl -I http://192.168.95.129'. The output of the command is displayed, showing the HTTP status code '200 OK' and several headers: 'Date: Wed, 02 Sep 2026 15:36:59 GMT', 'Server: Apache/2.2.8 (Ubuntu) DAV/2', 'X-Powered-By: PHP/5.2.4-2ubuntu5.10', and 'Content-Type: text/html'. The terminal background features a Kali Linux logo watermark. The window title bar indicates it's a VirtualBox session named 'kali-linux-2025.1'.

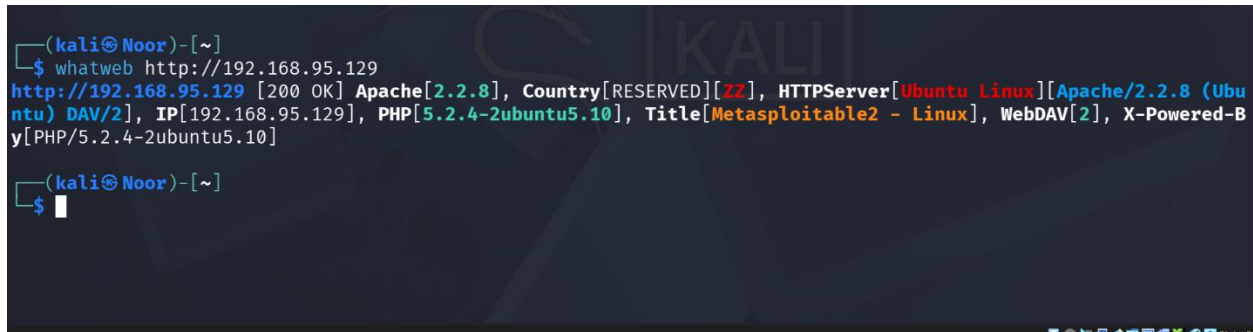
```
(kali@Noor)-[~]  
$ curl -I http://192.168.95.129  
HTTP/1.1 200 OK  
Date: Wed, 02 Sep 2026 15:36:59 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Type: text/html  
  
(kali@Noor)-[~]  
$
```

Status Code: HTTP/1.1 200 OK

Server Header: Server: Apache/2.2.8 (Ubuntu) DAV/2

Part 6 - WhatWeb Fingerprinting

Step 24: Basic fingerprint

A terminal window on a Kali Linux system. The prompt is (kali@Noor)-[~]. The user enters the command \$ whatweb http://192.168.95.129. The output is: http://192.168.95.129 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.95.129], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]. The prompt is then (kali@Noor)-[~] and the user enters a dollar sign \$.

```
(kali@Noor)-[~]  
$ whatweb http://192.168.95.129  
http://192.168.95.129 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.95.129], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
  
(kali@Noor)-[~]  
$
```

Record all recognisable technologies.

Web Server: Apache 2.2.8

Operating System: Ubuntu Linux

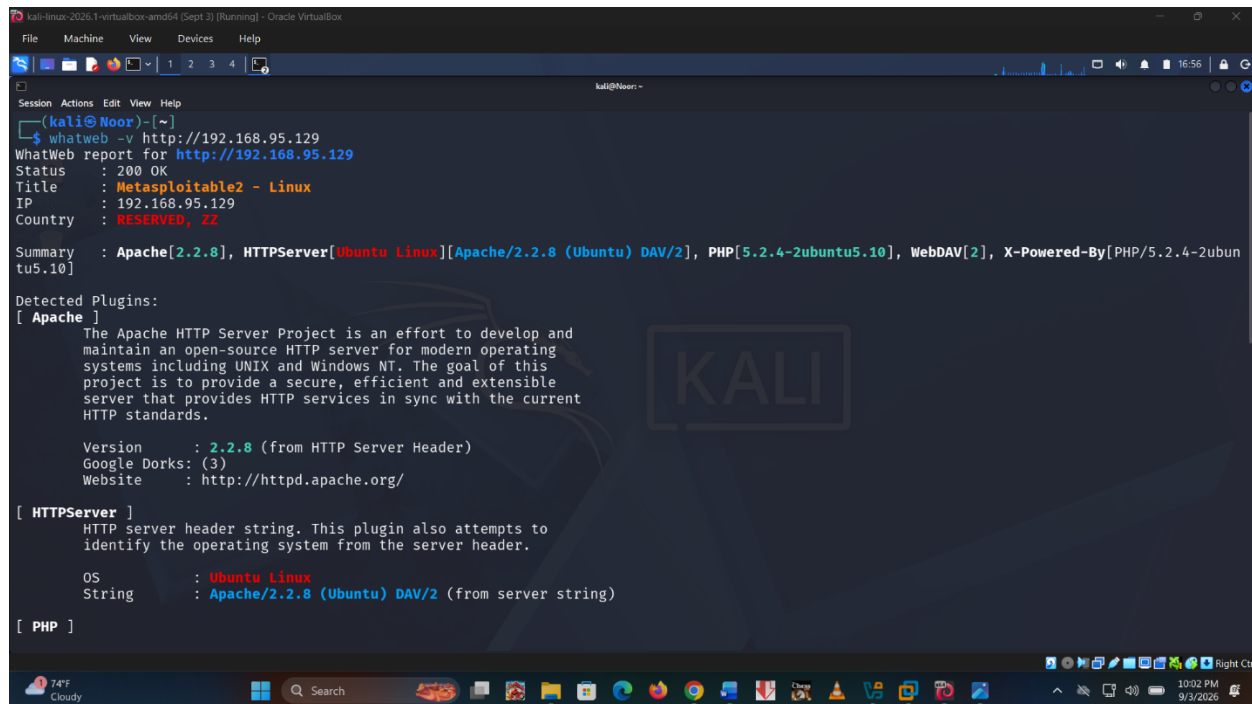
Server Extension Module: DAV 2 (WebDAV)

Programming Language Scripting Engine: PHP 5.2.4-2ubuntu5.10

HTML Page Title: Metasploitable2 - Linux

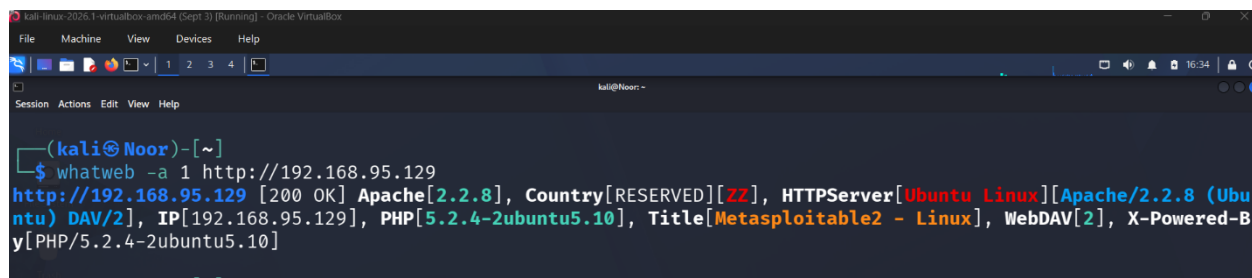
HTTP Status Code: 200 OK

Step 25: Verbose output



```
kali@Noor: ~  
$ whatweb -v http://192.168.95.129  
WhatWeb report for http://192.168.95.129  
Status : 200 OK  
Title : Metasploitable2 - Linux  
IP : 192.168.95.129  
Country : RESERVED, ZZ  
  
Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.  
Version : 2.2.8 (from HTTP Server Header)  
Google Dorks: (3)  
Website : http://httpd.apache.org/  
  
[ HTTPServer ]  
HTTP server header string. This plugin also attempts to identify the operating system from the server header.  
OS : Ubuntu Linux  
String : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)  
  
[ PHP ]
```

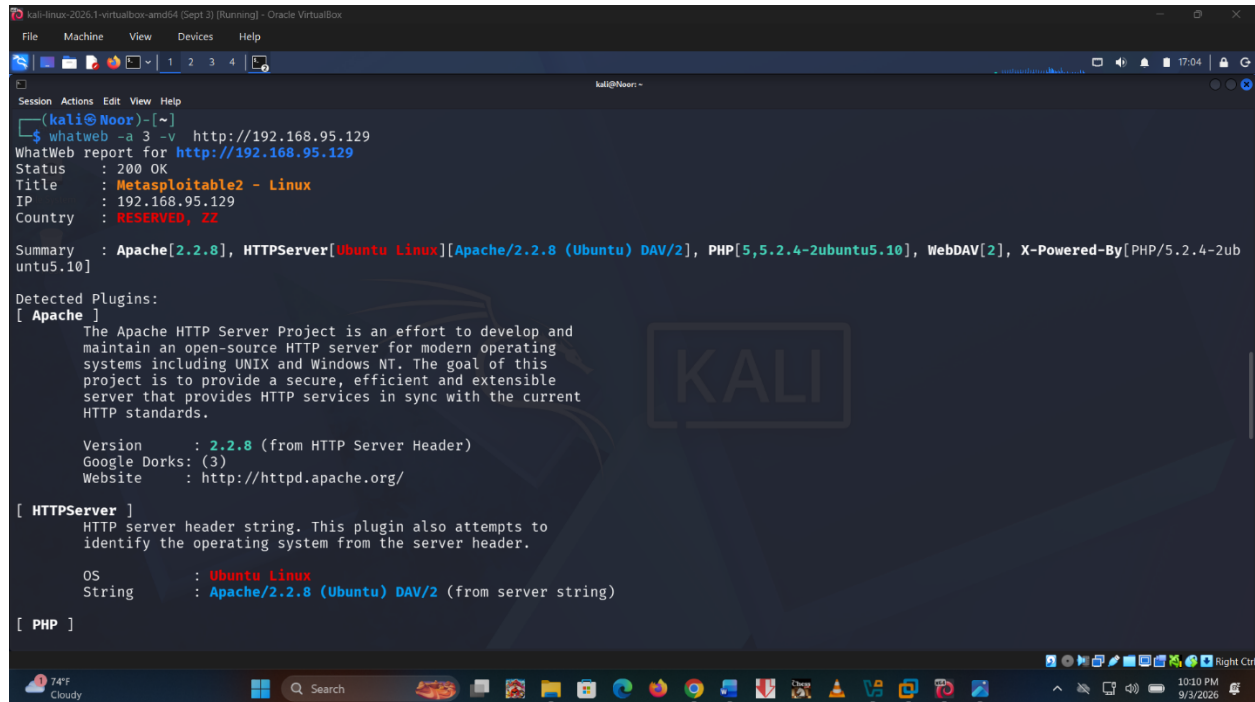
Step 26: Aggression level 1



```
kali@Noor: ~  
$ whatweb -a 1 http://192.168.95.129  
http://192.168.95.129 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.95.129], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

Both Step 24 and Step 26 give the same result as seen

Step 27: Aggression level 3

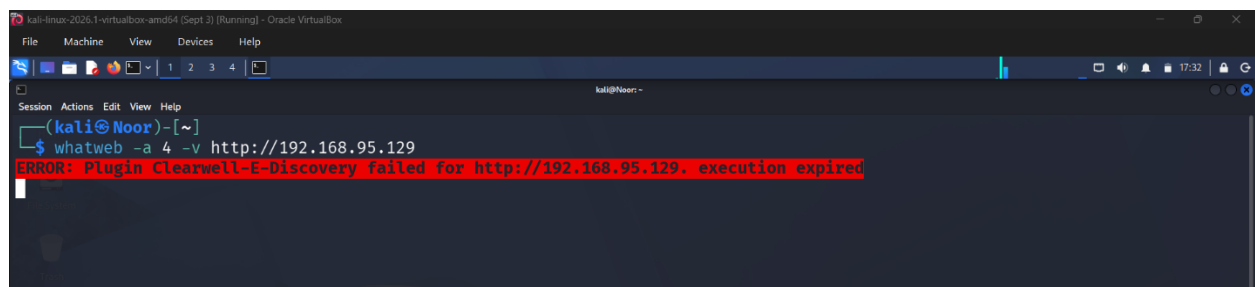


```
kali@Noor: ~  
$ whatweb -a 3 -v http://192.168.95.129  
WhatWeb report for http://192.168.95.129  
Status : 200 OK  
Title : Metasploitable2 - Linux  
IP : 192.168.95.129  
Country : RESERVED, ZZ  
  
Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.  
Version : 2.2.8 (from HTTP Server Header)  
Google Dorks: (3)  
Website : http://httpd.apache.org/  
  
[ HTTPServer ]  
HTTP server header string. This plugin also attempts to identify the operating system from the server header.  
OS : Ubuntu Linux  
String : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)  
  
[ PHP ]
```

Comparison between the two level 1 and 3. Is totally different because level 1 is a total summary while level 3 is expansion of every words within.

Step 28: Aggression level 4

This is what I ma receiving all the time when I run the command



```
kali@Noor: ~  
$ whatweb -a 4 -v http://192.168.95.129  
ERROR: Plugin Clearwell-E-Discovery failed for http://192.168.95.129. execution expired
```

Step 29: Follow redirects: Another error message as seen when running the command

```
(kali@Noor)-[~]
$ whatweb --follow-redirect=always http://192.168.95.129
ERROR Opening: http://192.168.95.129 - execution expired

(kali@Noor)-[~]
$ whatweb --follow-redirect=always http://192.168.95.129
ERROR Opening: http://192.168.95.129 - execution expired

(kali@Noor)-[~]
$
```

Step 30: Save results: This is the result shows after running the command.

```
kali-linux-2025.1-virtualbox-amd64 (Sept 3) [Running] - Oracle VM VirtualBox
File Machine View Devices Help
1 2 3 4
kali@Noor: ~
Session Actions Edit View Help
(kali@Noor)-[~]
$ whatweb -v http://192.168.95.129 > whatweb-results.txt
cat whatweb-results.txt

WhatWeb report for http://192.168.95.129
Status : 200 OK
Title : Metasploitable2 - Linux
IP : 192.168.95.129
Country : RESERVED, ZZ

Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

Version : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website : http://httpd.apache.org/
```

```
kali@kali:~$ curl -s -I http://10.10.10.10/
HTTP/1.1 200 OK
Date: Wed, 02 Sep 2026 17:57:43 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to
  identify the operating system from the server header.

  OS      : Ubuntu Linux
  String  : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
  PHP is a widely-used general-purpose scripting language
  that is especially suited for Web development and can be
  embedded into HTML. This plugin identifies PHP errors,
  modules and versions and extracts the local file path and
  username if present.

  Version   : 5.2.4-2ubuntu5.10
  Google Dorks: (3)
  Website   : http://www.php.net/

[ WebDAV ]
  Web-based Distributed Authoring and Versioning (WebDAV) is
  a set of methods based on the Hypertext Transfer Protocol
  (HTTP) that facilitates collaboration between users in
  editing and managing documents and files stored on World
  Wide Web servers. - More Info:
  http://en.wikipedia.org/wiki/WebDAV

[ X-Powered-By ]
  X-Powered-By HTTP header

  String    : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)

HTTP Headers:
HTTP/1.1 200 OK
Date: Wed, 02 Sep 2026 17:57:43 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html
```

Part 8 - Student Questions

1. What is reconnaissance?

Reconnaissance is the very first phase of a penetration test or cyberattack, focused on gathering as much information as possible about a target system or network.

2. What is the difference between host discovery and port scanning?

Host discovery finds out if a machine is alive, while port scanning finds out what services are running on that alive machine

3. Explain open, closed and filtered.

Open: An application is actively listening and accepting connections on this port. **These are your direct entry points for Metasploit exploits.**

Closed: The host is online, but no application is listening on this port. It is safe from attacks because there is no software to exploit.

Filtered: A firewall or network defense is actively blocking or dropping Nmap's packets. Nmap cannot tell if the port is open or closed because it receives total silence.

4. What does -sV do?

The **-sV** flag enables **Service Version Detection** in Nmap [Metasploit Unleashed - Requirements]

5. What does -O do?

The **-O** flag enables **Operating System (OS) Detection** in Nmap.

6. Explain what -A combines and why it is noisier.

The **-A** flag enables **Aggressive Scan** mode in Nmap. It is a powerful shortcut flag designed to maximize information gathering by consolidating several independent scanning tasks into a single execution.

7. What does -p- mean?

The **-p-** flag instructs Nmap to scan **every single possible TCP/UDP port (all 65,535 ports)** on the target machine.

8. Why can UDP scans take longer?

UDP scans take significantly longer than TCP scans primarily because of how the underlying protocol handles communications and how target operating systems respond to errors

9. What does -sC do?

The **-sC** flag instructs Nmap to run a scan using the default set of scripts from the Nmap Scripting Engine (NSE).

10. What information did the HTTP title/headers scripts reveal?

The HTTP title and headers Nmap scripts (http-title and http-headers) revealed **three critical layers of information** about the Metasploitable 2 web server running on port 80.

11. What did SMB/SSH/FTP enumeration add?

While HTTP enumeration (like whatweb or curl) maps out web services, **SMB, SSH, and FTP enumeration** added deep validation of systemic access rules. They bypassed web boundaries to expose misconfigurations and software signatures that grant **complete operating system control**.

12. What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap focuses on infrastructure and transport services (Layer 4), while WhatWeb focuses on web application software and frameworks (Layer 7).

13. How did WhatWeb levels 1, 3 and 4 differ?

Level 1 (Stealthy): Makes exactly **one HTTP request** per URL. It only reads information naturally sent back in the initial page response (like headers and page titles). It is very fast, completely silent, and looks like a normal user browsing the site.

Level 3 (Aggressive): Uses **conditional logic**. If Level 1 discovers a specific platform (like WordPress or Apache), Level 3 actively triggers specialized, secondary checks for that platform only. It makes a handful of extra requests, creating a medium amount of background noise.

Level 4 (Heavy): Fires **every single plugin in WhatWeb's database** to perform brute-force checks, regardless of what software is actually there. It floods the web server with thousands of requests to check for hidden panels and asset paths. This is extremely loud, triggers firewalls, and caused your execution expired timeout error on the lightweight Metasploitable 2 VM.

14. Why must aggressive scanning/fingerprinting remain inside the authorised lab?

Aggressive scanning, active fingerprinting, and heavy application probing must strictly remain inside an authorized lab environment for three primary reasons: **legal protection, network safety, and operational security**.